Multistatic Radar Simulator — System Architecture & Data Flow

E. Sai Shervan (CS24B073) · CS5013 · Packages under com.radar.simulator · same pipeline for the GUI and the CLI

1 · CONFIGURE

GUI form
`ui.ScenarioEditor`
TX, receivers, drone, timing

Command line
`CLISimulatorApp`
--receivers --duration --timestep

Saved scenario file
`persistence.ScenarioPersistence`
load .properties

Build & check input
`ui.ScenarioConfig.toScenario()`
numbers, unique IDs, timing

Scenario
`scenario.Scenario`
Transmitter + Receivers
+ TrajectoryModel

bad value

Invalid input
red message names the bad field
run is not started

2 · RUN THE EXPERIMENT

Geometry valid?
`localization.GeometryValidator`

no

Refused
explicit Status:
UNDER_CONSTRAINED
INVALID_INPUT
NUMERICALLY_UNSTABLE

yes (coplanar sensors → warning)

experiment.ExperimentRunner — repeat for every time step i (t = i · Δt)

① True drone position
`scenario.TrajectoryModel.getPosition(t)`
constant velocity: p(t) = p_0 + v · t

② Simulate measurements
`measurement.MeasurementModel.generate()`
bistatic range = |p − TX| + |p − RX|, delay = range / c, power
one ReceiverMeasurement per receiver, keyed by receiver ID

③ Estimate position
`localization.LocalizationModel.estimate()`
Levenberg–Marquardt least squares on the range ellipsoids
→ EstimationResult: position, Status, residual, condition no.

④ Compare with the truth
error = |true position − estimate|
(NaN if Status ≠ SUCCESS, excluded from metrics)

⑤ Record the step
`experiment.ExperimentRecord`
time, truth, measurement, estimate, error,
measurement + localization model name/version

next time step

Evaluate the run
`experiment.EvaluationModule → EvaluationResult`
mean / max / RMS error, localized vs failed steps

3 · SHOW & SAVE RESULTS

shown in red

GUI (ui.SimulatorController)
Visualization3D: sensors, true path, estimates, playback
ResultsPanel: per-step table + mean / max / RMS
Status line: model versions, warnings, errors
`SceneTransform: East/North/Up → JavaFX axes`

CLI output
per-step table + evaluation summary in the terminal

Save / export
`persistence.ScenarioPersistence`
save scenario (.properties) · export results CSV
+ evaluation summary (.txt)

Replaceable formulations (interfaces)

`TrajectoryModel`	← ConstantVelocityTrajectory
`MeasurementModel`	← BaselineMeasurementModel v1.0
`LocalizationModel`	← BaselineLocalizationModel v2.0

The stakeholder's formulation plugs in as a new class
with its own name/version — the rest of the pipeline is unchanged.

Package colours: ui · scenario · measurement · localization · experiment · persistence · errors / refused

dashed arrows = error path · ①–⑤ run once per time step